

Figure S3 Candidate program-level targets from the two-axis reprogramming

Candidate target	Level	Rewired program	Evidence (bulk n=4 vs 4)	Cross-species	Intervention
PPARa	nuclear receptor	FAO / FA-transport (down)	Ppara -2.45 (***)	FAO conserved	agonist (restore oxidation) — but worsened sur
FXR	nuclear receptor	biliary axis (shut down)	Nr1h4 -1.68 (***)	biliary conserved	agonist (anti-cholestatic)
ABCA1	effector	HDL efflux (induced)	Abca1 +3.00 (***)	only program not rebounding	enhance (LPS clearance)
FASN	effector	lipogenesis (induced)	Fasn +2.19 (***)	discordant (human vs mouse)	deferred (time-resolved study needed)
CD36	effector	scavenger SR-B (induced)	Cd36 +0.76 (ns)	not conserved	deferred (direction not conserved)

Evidence = independent purified-hepatocyte bulk RNA-seq (GSE311736, n = 4 vs 4, DESeq2); *, **, *** = padj < 0.05/0.01/0.001; ns = not significant.
PPARa/FXR are the two suppressed upstream nuclear receptors; ABCA1 induction is dissociated from LXRα transcript abundance (Fig 4); the upstream driver is not identified and was not tested here.